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A Multi-Stage Efficiency Analysis of Taiwan's Semiconductor Industry: Integrating ESG and Artificial Intelligence Perspectives

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ABSTRACT

The semiconductor industry is central to global technological progress but increasingly faces the challenge of balancing financial performance with rising environmental, social, and governance (ESG) expectations. While ESG and artificial intelligence (AI) are widely regarded as strategic priorities, limited empirical research has investigated their combined influence on firm-level efficiency in high-tech, capital-intensive sectors. This study addresses this gap by applying a three-stage Data Envelopment Analysis (DEA) to evaluate the performance of 33 publicly listed semiconductor firms in Taiwan across profitability, sustainability, and market value. The model reflects the sequential value-creation process: the Profit stage measures how R&D and labor are transformed into innovation output (intangible assets and income); the Sustainability stage assesses how governance and environmental inputs shape ESG scores; and the Market stage examines how ESG performance and capital investment affect market valuation. Stage-specific efficiency is measured using ESG and structural variables aligned with the semiconductor sector's vertically disintegrated and interdependent value chain. AI-driven innovation is proxied by patent activity. Results reveal performance disparities across stages: Market stage efficiency ranks highest (0.9664), followed by Profit stage, while Sustainability stage lags—highlighting an ESG implementation gap in otherwise competitive firms. Tobit regression analysis shows that AI-driven innovation enhances sustainability efficiency, whereas firm size and leverage are more closely linked to profitability. This study proposes a multi-stage framework integrating ESG and AI to evaluate sustainable innovation in high-tech sectors, offering insights into how firms can balance competitiveness with long-term sustainability.

1 | Introduction

The global semiconductor industry, valued at approximately USD 503 billion in 2023 and projected to reach USD 980 billion by 2029 (Statista 2023), constitutes a foundational pillar of technological progress in domains such as artificial intelligence (AI), computing, and communications. Taiwan's central position within this global ecosystem is underscored by its substantial contributions, accounting for 77.9% of global wafer foundry output, 21.3% of integrated circuit (IC) design, and 52%

of IC packaging and testing. Supported by leading firms such as Taiwan Semiconductor Manufacturing Company (TSMC) and MediaTek, Taiwan's semiconductor sector exhibits resilience and global competitiveness amid escalating geopolitical tensions and supply chain vulnerabilities. Nevertheless, the sector's rapid expansion has exacerbated its environmental footprint and energy intensity, underscoring the need to implement ESG frameworks to mitigate risks associated with unsustainable growth. Taiwan's distinctive position—marked by advanced technological capabilities, high-value innovation output, and a

central role in the global supply chain—renders it both a global leader and a focal point for ESG accountability. Against the backdrop of intensifying international and domestic demands for corporate sustainability, the industry faces mounting pressure to align with environmental, social, and governance (ESG) standards, necessitating a strategic equilibrium between global competitiveness and sustainable development.

Recent policy developments, including Taiwan's Climate Change Response Act and the upcoming 2025 carbon fee, underscore the urgency of adopting more integrated ESG strategies within the sector. Nonetheless, the implementation of ESG principles remains inconsistent across firms, particularly in relation to standardized sustainability reporting, carbon emissions mitigation, and supply-chain transparency. For small and medium-sized enterprises (SMEs), the adoption of ESG frameworks often lags due to limited resources and technical capacity. Despite these challenges, the potential of AI technologies to support ESG performance—through applications such as real-time emissions monitoring, energy optimization, and automated ESG scoring—has been increasingly recognized (Ozkan 2024; Svensson 2022).

While this rapid growth highlights the industry's importance, it also magnifies its environmental footprint and governance complexity, particularly in resource- and energy-intensive production processes. The sector's decarbonization demands have shifted ESG integration from a reputational concern into a strategic imperative. Nevertheless, empirical research remains limited on how ESG strategies and AI innovation can be jointly leveraged to improve performance in high-impact industries. Consequently, there is a critical need to develop analytical frameworks that capture how firms operationalize sustainability alongside technological advancement and how these efforts influence multidimensional efficiency.

Although the theoretical complementarity between AI and ESG has been conceptually acknowledged, empirical studies examining their joint impact on firm performance remain scarce. Existing research often evaluates ESG outcomes in isolation from efficiency metrics or focuses narrowly on environmental disclosures, without accounting for technological enablers or industry-specific constraints. Moreover, there is limited understanding of how ESG and AI interact within capital- and resource-intensive industries such as semiconductors, where sustainability goals may conflict with high-performance production demands.

This study addresses these gaps by examining how ESG practices and AI-driven innovation jointly influence firm-level efficiency in Taiwan's semiconductor industry. Taiwan provides an ideal case study due to its global dominance in wafer foundry, IC design, packaging, and a vertically integrated supply chain, as well as advanced technological capabilities, rendering it highly competitive and subject to heightened ESG scrutiny. To capture this complexity, we apply a three-dimensional efficiency framework based on Data Envelopment Analysis (DEA), which evaluates firm performance across the Profit stage, Sustainability stage, and Market stage. DEA is particularly suitable in this context as it simultaneously accounts for multiple inputs and outputs. It offers a robust analytical tool to assess the interplay

between innovation and ESG practices within a capital- and energy-intensive sector.

Building on the DEA results, a Tobit regression model is employed to identify the determinants of efficiency across the three dimensions, with a specific focus on the enabling role of AI. The Tobit model is appropriate given that efficiency scores are bounded between 0 and 1, and it yields statistically consistent estimates that ordinary least squares (OLS) cannot provide. This integrated approach enables the linkage of operational assessment with explanatory analysis, thereby elucidating the mechanisms through which AI-driven innovation and firm-specific attributes shape efficiency trajectories.

This study makes three key contributions: (1) it introduces an innovative three-stage efficiency analysis framework that integrates ESG dimensions to evaluate corporate performance comprehensively; (2) it quantifies the impact of AI-driven innovation, measured through patent activity, on efficiency using Tobit regression analysis; and (3) it provides actionable insights into the characteristics of high-, medium-, and low-efficiency firms, offering evidence-based recommendations for achieving the United Nations' (UN) Sustainable Development Goals (SDGs) and enhancing market competitiveness. These findings contribute to the broader understanding of how ESG and AI integration can drive efficiency and sustainability in a globally critical industry.

The remainder of the paper is structured as follows: Section 2 reviews the relevant literature, Section 3 outlines the methodology, Section 4 presents the results, and Section 5 concludes with key findings and implications.

2 | Literature Review

2.1 | ESG and Corporate Efficiency: A Multi-Dimensional Perspective

Although prior research has examined ESG performance and AI capabilities as separate constructs, few studies have investigated their synergistic impact on corporate efficiency within capital- and energy-intensive industries (Di Vaio et al. 2020). Existing scholarship has largely focused on overarching sustainability frameworks and the SDGs, with limited attention to the direct empirical relationship between ESG adoption and firm-level efficiency outcomes (Singh et al. 2024). This study addresses this gap by integrating ESG and AI perspectives within a multi-stage efficiency evaluation framework tailored to the semiconductor sector. The ESG framework provides a comprehensive lens for assessing corporate efficiency beyond traditional financial metrics, encompassing environmental indicators such as carbon emissions and energy consumption, social dimensions including labor conditions and workforce diversity, and governance attributes such as board independence and disclosure transparency. These ESG components directly influence firms' resilience, innovation capacity, and long-term competitiveness, particularly in resource-intensive sectors like semiconductors (Chen and Delmas 2011; Li et al. 2021). Accordingly, this section delineates the operational relevance of each ESG pillar and its implications for semiconductor efficiency performance.

The environmental dimension is particularly salient due to the high carbon emissions, energy consumption, and water usage inherent in semiconductor manufacturing. Agrawal (2023) identified that inefficiencies in semiconductor wafer fabrication—particularly equipment underutilization and scheduling fragmentation—are closely linked to increased energy consumption and operational costs, emphasizing the need for process optimization. Metrics such as the Environmental score (E-score) assess environmental practices, including emissions reduction and renewable energy adoption. Empirical evidence suggests that firms with higher E-scores often exhibit superior operational efficiency (Chen and Delmas 2011).

In Taiwan, regulatory developments such as the 2023 Climate Change Response Act and the forthcoming 2025 carbon fee have significantly elevated the strategic importance of environmental performance. These policies incentivize the adoption of cleaner technologies and ESG reporting systems (Sueyoshi and Ryu 2020), reinforcing the role of traceability and transparency in low-carbon transitions (De Villiers et al. 2021). Recent reviews by Di Vaio et al. (2025) and Di Vaio and Ali (2025) further highlight the critical role of disclosure and structured accountability—particularly relevant in emission-intensive sectors like semiconductors.

Beyond environmental sustainability, the social dimension is essential for maintaining organizational efficiency through workforce stability, innovation capacity, and reputation. Indicators such as labor relations, diversity, and workplace safety directly affect productivity and long-term human capital development. Lu et al. (2013) found that US semiconductor firms engaging in socially responsible practices achieved higher operational performance. In Taiwan, growing public scrutiny of working conditions and gender equity in STEM fields has further heightened expectations for social responsibility. The Social score (S-score) captures factors such as employee rights, equal opportunity, and supply chain responsibility, reflecting a firm's ethical standards and operational resilience.

Complementing the environmental and social pillars, the governance dimension serves as the institutional backbone for ESG implementation. Effective governance structures—characterized by board independence, diversity, and audit transparency—enhance oversight, strategic coherence, and capital efficiency. Strong governance also reduces agency costs and strengthens long-term sustainability orientation. Li et al. (2021) demonstrated that robust governance practices improve ESG alignment and performance outcomes. Moreover, sector-specific ESG research underscores the need for customized metrics and governance frameworks, particularly in high-technology sectors where general indicators may fail to capture process-specific nuances (Delmas and Toffel 2014; Baumgartner 2014).

2.2 | AI and Corporate Efficiency: Opportunities and Challenges

Building on the ESG framework discussed above, this section examines how AI influences corporate efficiency—both as a technological enabler and a potential disruptor. “AI-driven innovation” refers to firm-level technological advancements enabled

by AI systems, particularly those that enhance process control, product yield, and sustainability analytics in semiconductor manufacturing. Examples include machine learning models for equipment optimization, defect detection algorithms, and automated ESG reporting tools. This conceptualization aligns with the operational characteristics of innovation in high-tech manufacturing, where AI functions as both a production enabler and a decision-support mechanism.

AI has become a transformative driver of corporate efficiency, with wide-ranging applications across productivity optimization, innovation enhancement, and sustainability integration. AI applications in firms can be broadly categorized into operational optimization, ESG performance evaluation, and strategic innovation (Zhang et al. 2023). In the semiconductor industry—marked by rapid technological cycles and complex manufacturing demands—AI is increasingly leveraged for predictive analytics, process automation, and real-time decision support (Shi et al. 2023). However, the efficiency implications of AI adoption are multifaceted, and recent studies call for more balanced evaluations that account for both opportunities and systemic risks.

Empirical evidence shows that AI enhances firm efficiency by improving throughput and optimizing resource allocation. In China's manufacturing sector, Luo et al. (2024) found that AI-based technological innovation significantly improved total factor productivity, particularly in regions with strong digital infrastructure and marketization, underscoring the role of enabling environments. In ESG contexts, AI supports emissions tracking, energy management, and ESG rating analytics (Capelle-Blancard and Petit 2022; Pan et al. 2024), aligning with Di Vaio et al. (2020), who emphasized AI's role in sustainable business models aligned with the UN's SDGs. AI also contributes to green manufacturing through real-time monitoring, process optimization, and regulatory compliance.

Di Vaio et al. (2020) highlighted AI's dual role in process innovation and sustainability integration, especially in high-tech sectors. Frank (2021) noted that AI-enabled ESG practices vary by geography and product, while Liu et al. (2022) found that AI significantly improved energy efficiency in China. In the semiconductor context, Tassel et al. (2023) showed that self-supervised learning optimized fab scheduling, reducing energy use and boosting throughput—demonstrating AI's transformative potential within the industry.

In Taiwan's semiconductor sector, AI-driven innovation is proxied by patent activity, indicating firms' capabilities in automation and advanced production. Firms with higher AI intensity exhibit stronger innovation and competitiveness (Rammer et al. 2021), and AI complements existing IT infrastructure to enhance efficiency (Zhang et al. 2023). Applications include real-time monitoring, fab scheduling, energy optimization, and predictive maintenance (Tassel et al. 2023), all of which contribute to improved operational and environmental performance.

However, notable operational and ethical risks remain. While AI holds promise for enhancing ESG performance, its deployment may also introduce unintended consequences that threaten long-term sustainability and social equity. Huang (2024) cautioned

that AI adoption may exacerbate labor market inequality by disproportionately displacing mid-skilled occupations that are vulnerable to automation. Singh et al. (2024) highlighted ethical and inclusiveness concerns, while Larsson and Heintz (2020) pointed to algorithmic opacity and bias. These risks are particularly salient in technology-intensive sectors, where AI adoption may outpace regulatory and organizational safeguards. For example, in Taiwan, rising AI adoption has increased demand for algorithm engineers while leaving mid-skilled technician roles stagnant, creating reskilling challenges (Yang 2022).

Nevertheless, when aligned with sustainability goals and governed by ethical frameworks, AI can serve as a catalyst for ESG-oriented innovation (Ozkan 2024). These dynamics underscore AI's dual nature—both an enabler and a disruptor—requiring firms to develop strategic governance mechanisms that ensure responsible deployment and accountability.

2.3 | ESG, AI, and the Semiconductor Industry: Toward an Integrated Framework

To synthesize the preceding discussions on ESG and AI, this section proposes an integrated conceptual framework tailored to the semiconductor industry. Based on the creating shared value (CSV) perspective (Porter and Kramer 2011), sustainability is framed as a compliance requirement and a strategic driver of competitive advantage. Unlike traditional corporate social responsibility (CSR), which often emphasizes compliance or philanthropy, CSV embeds environmental and social objectives directly into the value-creation process, positioning them as sources of productivity and innovation. Within this framework, environmental and social goals are aligned with firm-level efficiency and innovation, increasingly facilitated by AI applications that support real-time optimization, traceability, and data-driven decision-making.

Although ESG and AI have each been widely studied as independent performance drivers, few empirical investigations have examined their combined effects on firm-level efficiency, particularly in capital- and energy-intensive industries such as semiconductors. This research gap is especially pronounced in Taiwan, where semiconductor firms operate within globally integrated supply chains and face intensifying regulatory, environmental, and geopolitical challenges. A sector-specific framework is therefore essential to capture the interplay between ESG implementation, AI-driven innovation, and performance outcomes.

The semiconductor industry presents unique ESG-related challenges. From an environmental perspective, chip fabrication requires ultrapure water, substantial energy input, and the management of hazardous substances, making resource efficiency and decarbonization both urgent and difficult to achieve. Socially, the sector is heavily dependent on skilled labor yet faces persistent concerns regarding worker burnout, gender disparities, and labor rights—issues particularly salient in Taiwan's export-oriented, high-tech economy. Regarding governance, concentrated ownership structures and family-controlled boards may impede transparency unless formal oversight mechanisms are instituted (Lu et al. 2013). Recent research (Sai and

Yamada 2025) suggests that passive ownership is associated with stronger board independence, greater diversity among outside directors, and improved disclosure, whereas concentrated control may erode stakeholder trust. In semiconductor M&A, tightening regulatory scrutiny—especially in cross-border deals—underscores the need for governance safeguards to mitigate information asymmetries and ensure ESG accountability.

These sector-specific dynamics highlight the need for tailored ESG metrics and implementation frameworks. For example, TSMC's ESG scorecard includes industry-calibrated indicators such as water consumption per wafer and greenhouse gas emissions per process node, which are embedded within supplier audit systems. Such practices illustrate a shift toward contextualized ESG benchmarking in the semiconductor sector.

However, ESG benchmarking alone is insufficient. In the absence of real-time data and adaptive modeling, firms risk stagnation or superficial compliance. AI offers a powerful complement to ESG strategy, enabling tools such as emissions forecasting, anomaly detection, unstructured ESG data extraction, and risk modeling—thereby facilitating dynamic, large-scale monitoring and execution (Capelle-Blancard and Petit 2022; Pan et al. 2024).

The theoretical basis for this convergence is grounded in CSV and integrated reporting frameworks (Eccles et al. 2012), which emphasize value co-creation, transparency, and long-term accountability. Furthermore, AI-enhanced ESG systems align with stakeholder theory, which positions firms as accountable not only to shareholders but also to a broader network of stakeholders. In this light, AI functions both as a technological enabler and a governance mechanism capable of reinforcing ESG integration.

Recent regulatory developments—such as Taiwan's 2023 Climate Change Response Act and the impending 2025 carbon fee—underscore the urgency for ESG–AI alignment. These instruments impose financial accountability for emissions and incentivize the adoption of AI-enabled compliance tools for carbon risk modeling, energy optimization, and ESG reporting. The coevolution of digital technologies and sustainability regulation is thus reshaping ESG capabilities in high-tech industries.

Nevertheless, integration is not without risks. Issues related to equity, transparency, and algorithmic accountability persist. Scholars caution that unchecked AI adoption may reinforce labor displacement, amplify bias, and reduce interpretability in decision systems (Singh et al. 2024; Huang 2024). As such, effective ESG–AI integration necessitates robust ethical oversight and institutional safeguards.

This study proposes a three-dimensional efficiency evaluation model that integrates ESG indicators, AI-driven innovation (proxied by patent activity), and firm-specific characteristics to address these conceptual and empirical gaps. Patent activity is adopted as a proxy for AI-driven innovation, as it reflects a firm's capacity to codify, protect, and commercialize technological advancements—particularly those about automation, predictive analytics, and process optimization, which are foundational to AI deployment in semiconductor manufacturing. This model captures both the reinforcing synergies

and mutual constraints between sustainability strategies and technological capabilities, offering a comprehensive framework to assess performance in the semiconductor industry. While ESG and AI have been extensively examined as separate determinants of firm performance, their joint influence on efficiency—especially in capital- and knowledge-intensive sectors—remains underexplored. This study aims to bridge this gap by proposing an integrated ESG–AI efficiency model tailored to the semiconductor industry.

3 | Data and Methodology

3.1 | Data Sources and Variable Descriptions

This study evaluates the efficiency of 33 domestically listed semiconductor companies, as detailed in Table 1, which provides information on the proportions of their main products (as of December 2022). Each company is treated as a decision-making unit (DMU). Financial statement data spanning 5 years (January 1, 2018, to December 31, 2022) was obtained from the Taiwan Economic Journal (TEJ) database.

This study adopts a two-phase analytical framework, applying DEA in the first stage and Tobit regression in the second. First, a dynamic three-stage network Data Envelopment Analysis (DEA) model is employed to assess firm performance across three interrelated dimensions: profitability, sustainability, and market value. Based on Tone and Tsutsui (2010), this model incorporates intermediate outputs and carry-over variables, making it well-suited to the temporal and structural characteristics of innovation and ESG processes. The vertically disintegrated and innovation-intensive nature of Taiwan's semiconductor industry further supports the model's relevance, as demonstrated in previous DEA applications to the sector (Lu and Hung 2010; Lu et al. 2022). Similar multi-stage DEA structures have also been effectively applied in other manufacturing sectors, underscoring their empirical robustness for capturing dynamic and multidimensional performance (Azadi, Moghaddas, et al. 2023; Azadi, Yousefi, et al. 2023).

Second, a Tobit model analyzes how firm-specific attributes—such as innovation output, size, and capital structure—affect stage-specific efficiency scores. Given that the DEA scores are bounded between 0 and 1, the Tobit model offers consistent parameter estimates, unlike ordinary least squares (OLS) regression.

Combining DEA with Tobit analysis facilitates a two-stage explanatory framework. Efficiency scores from each DEA stage serve as dependent variables, enabling the identification of structural drivers behind performance differences. This design bridges operational assessment with strategic interpretation by linking firm capabilities in innovation, ESG engagement, and capital decisions to their efficiency trajectories.

The DEA model decomposes firm performance into three sequential stages over time, as shown in Figure 1. In Stage 1 (Profit), inputs include R&D expenditure and the number of employees, capturing technological and labor resources (Fried

et al. 2008). Outputs—operating income and intangible assets—reflect returns on innovative investment. Operating income also serves as an intermediate output to Stage 2 (Sustainability), transmitting operational efficiency to ESG performance (Tone and Tsutsui 2010).

Stage 2 (Sustainability) focuses on ESG efficiency. Inputs such as executive compensation, employee benefits, and greenhouse gas scores represent governance, labor welfare, and environmental responsibility (Li et al. 2019). Outputs include E-score (environmental), S-score (social), and G-score (governance), which feed into Stage 3, linking sustainability practices to stakeholder perception and firm value.

Stage 3 (Market) uses market value as the sole output, reflecting overall investor confidence and long-term valuation performance. Fixed assets are introduced as a carry-over variable from period t to $t + 1$, capturing the lasting effect of capital investment and aligning with the logic of multi-period DEA (Tone and Tsutsui 2010; Lu and Hung 2010; Lu et al. 2022).

Figure 1 depicts this dynamic structure, where stages operate within each period but are interlinked through intermediate and carry-over variables. The model represents a continuous process of value creation, integrating operational, sustainability, and market-based perspectives.

This integrated framework provides a conceptually rigorous and empirically grounded platform to assess how ESG engagement and innovation capabilities jointly influence the efficiency trajectories of semiconductor firms—particularly in an industry where intangible assets and long-term strategies drive competitiveness.

The second part of this study uses regression analysis to examine the relationship between company characteristics and efficiency at each stage, investigating the mechanisms and extent of these variables' impacts. As a core indicator of innovation capabilities, the number of patents is crucial in evaluating technological innovation. Svensson (2022) highlights that the number of patents effectively measures innovation performance and technological innovation quality, particularly regarding its impact on subsequent technological development.

Company size is another key characteristic affecting corporate performance. Ahmed et al. (2023) identify a significant relationship between company size, ESG performance, and financial performance, noting that larger companies often invest more in environmental and social responsibility initiatives and have competitive advantages in the market. Parrao et al. (2024) emphasize that company size significantly influences innovation sustainability and job growth, particularly in resource-rich emerging markets. Thus, company size is included in the regression model to analyze its role in efficiency performance across stages.

Company age and internationalization are also important variables. Clauss et al. (2019) argue that company age affects environmental dynamism and innovation differently, with younger companies being more innovative while mature companies excel in resource integration and stability. Similarly, Mallingu

TABLE 1 | Semiconductor companies and their major product weight.

Codes/company abbreviations	Major product weight (2022/12)
2303 UMC	Consumer ICs (100%)
2330 TSMC	Integrated circuit chips (99.6%), others (0.4%)
2337 Macronix (MXIC)	ICs, smart products and others (100%)
2344 Winbond (WB)	AC/DC power management ICs (41.7%), electronic component distribution (33.9%), smart application (non-power) ICs (24.4%)
2351 SDI	Multimedia chips and mobile phone chips, etc. (97.9%), others (2.2%)
2363 SIS	Laptop input device modules (73%), consumer touch ICs (20.6%), others (5.7%)
2379 Reltek (RT)	Integrated circuits (ICs) (100%)
2388 VIA	Flat panel display driver chips (62.1%), SoC chips (37.1%), others (0.8%)
2408 Nanya technology (Nanya)	Application-specific ICs (55.7%), wafers (18.2%), commissioned design (14.4%)
2436 Weltrend	General ICs (91.6%), wafer foundry (6.9%), others (1.5%)
2449 KYEC	IC products and others (100%)
2454 Mediatek (MTK)	WiFi products (93.3%), others (6.7%)
2458 ELAN	Integrated circuit design (96.7%), others (3.3%)
3006 ESMT	Power management ICs (80.6%), temperature sensing ICs (8.5%), operational amplifier ICs (7%)
3034 Novatek	Wafer manufacturing integration (100%)
3035 Faraday	Wafers (87.1%), others (12.9%)
3227 PXI	Flash memory products (61.2%), ROM (32.7%), foundry (PC) (5.9%)
3529 eMemory	Logic products (46.5%), FLASH products (32.1%), DRAM products (18.7%)
4919 Nuvotane (NTC)	Electronics (85.6%), stationery (13%), others (1.3%)
4951 Fintek	IC testing (49.4%), wafer testing (33.2%), others (17.4%)
4952 Generalplus	CMOS image sensor components, bluetooth audio chips, etc. (97.4%), others (2.6%)
4968 RichWave	Technology royalties (69.7%), technical services (30.4%)
5274 Aspeed	Touch panel controllers/control chips (94.6%), audio/voice control chip modules (5.4%)
5299 EMC	Mixed-signal ICs (72%), power and charging management ICs (28%)
5347 VIS	Multimedia ICs (88.1%), computer peripheral ICs (10.5%), others (1.4%)
5351 Etron	Power components (78.7%), power management ICs (21.3%)
6147 Chipbond	Product sales (97%), others (3%)
6233 Prolific	Memory and analog logic system ICs (100%)
6457 Hycon	Digital system application products (63.7%), MEMS integrated application products (36.2%), others (0.1%)
6488 GWC	IC packaging (64.7%), IC testing (14.4%), module processing (8.5%)
6548 CWTC	Battery management chip product line (37.5%), mixed-signal microcontroller product line (28.5%), communication-related chip product line (25.8%)
8016 Sitronix	Semiconductor chips (97.4%), semiconductor ingot products (1.8%), others (0.7%)
8081 GMT	Leadframes for top-leaded ICs (75.4%), encapsulation resins (15.8%), LED leadframes (6.2%)

Source: Taiwan Economic Journal (TEJ) database; compiled by this study.

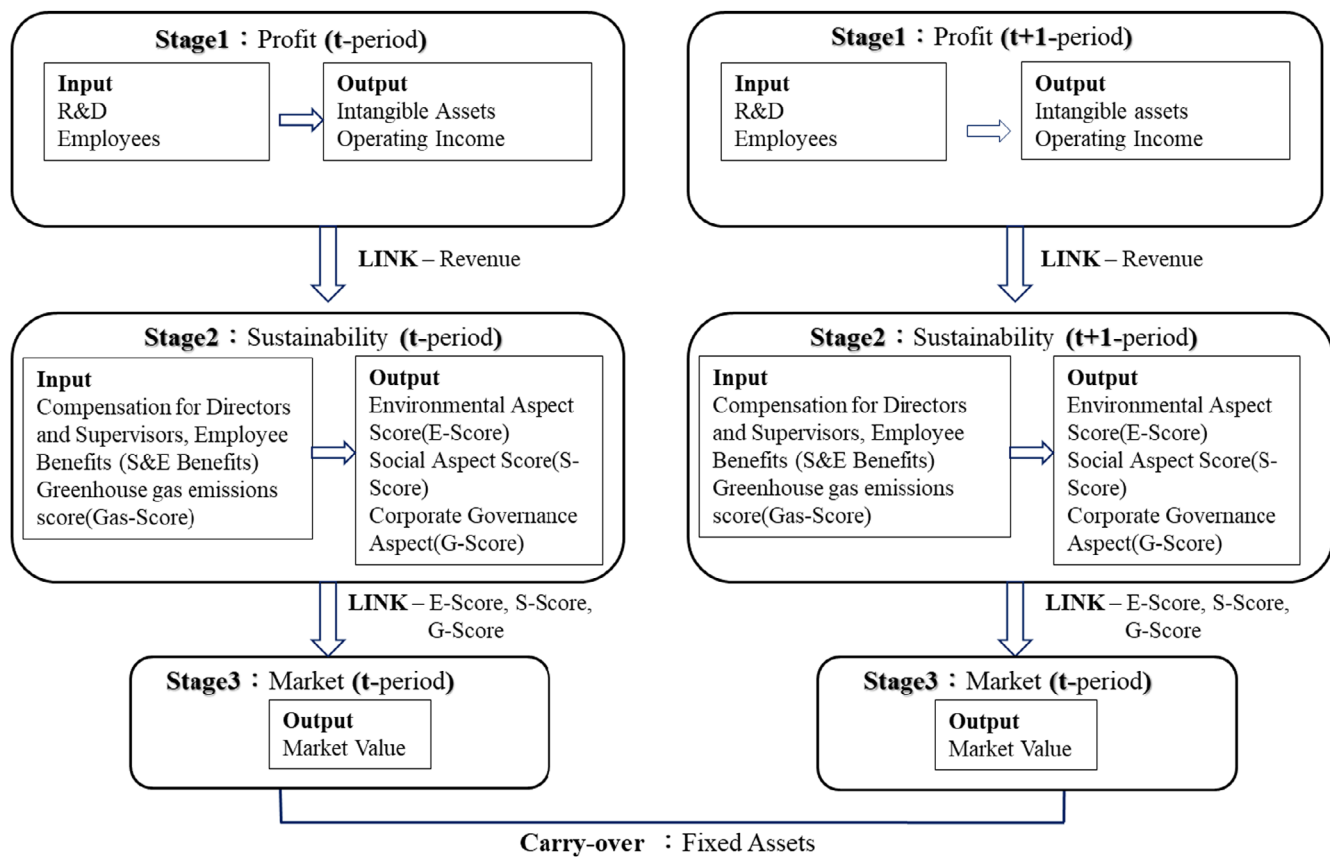


FIGURE 1 | Three-stage DEA model. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

et al. (2020) show that internationalization strategies significantly impact company performance. Based on these findings, this study examines the effects of company age and internationalization on efficiency performance in the Sustainability and Market stages.

Capital intensity is inversely related to corporate governance and information disclosure practices. As India shifts from traditional to research- and innovation-driven industries, the proportion of intangible assets in Indian companies is expected to grow. Ahmed et al. (2023) also note that the debt ratio reflects financial risk and capital structure efficiency, with moderate debt reducing agency costs and enhancing performance.

Traditional capital structure theory posits that the optimal capital mix minimizes the weighted average cost of capital to maximize market value. However, leverage and equity ratios alone cannot determine performance because of various interfering factors (Vätavu 2015).

This study incorporates firm-specific variables—including innovation output, enterprise scale, company age, capital intensity, and financial leverage—to examine their respective effects on efficiency across the Profit, Sustainability, and Market stages. These variables are theoretically grounded in prior studies and operationalized using standardized financial and ESG indicators. Their definitions, measurement units, theoretical roles, data sources, and supporting references

are comprehensively presented in Table 2, which aligns each variable with its corresponding stage in the three-stage DEA framework.

3.2 | Efficiency Analysis: DEA Model

This study employs a dynamic three-stage directional distance function (DDF) DEA model, assuming constant returns to scale (CRS). The model is defined as a set (x, y) , where x represents an input vector with m elements and y represents an output vector with s elements. The production possibility set $L(y)$ is defined as $L(y) = \{(x, y) \mid x \text{ can produce } y\}$, which represents all feasible input–output combinations.

Considering the direction vector $(x, y) \mid (g^x, g^y)$ for each input and output, Chung et al. (1997) proposed the concept of the DDF to measure the potential for input reduction and output expansion in a specific direction. It is defined as:

$$D(x, y, g^x, g^y) = \sup_{\alpha} \{ \alpha : (x - \alpha g^x, y + \alpha g^y) \in L(y) \},$$

where (g^x, g^y) is the direction vector indicating the adjustment direction of each input and output, expressed as $(g_j^x, g_j^y) = (g_{1j}^x, \dots, g_{mj}^x, g_{1j}^y, \dots, g_{sj}^y)$. (x_j, y_j) represents the input and output of the j th decision-making unit (DMU). The

TABLE 2 | Variable definition.

Stage	Factor	Definition	Numerical forms and units	Theoretical role/expected directional effect	Data source	Supporting references
<i>DEA model variables</i>						
Stage 1: Profit stage	R&D expenses (R&D)	Costs for research and development activities, covering personnel, facilities, materials, and other related expenditures	Number (thousand NT\$)	Input	Taiwan Economic Journal (TEJ)	Lu and Hung (2010); Lu et al. (2022); Chen and Yan (2011); Badiezadeh et al. (2018)
	Employees	Total personnel count including executives, managers, R&D staff, sales, and manufacturing	Number	Input	Taiwan Economic Journal (TEJ)	Zbranek (2013); Fried et al. (2008)
	Revenue (Rev)	Income from regular business operations, excluding sales allowances and returns. Identifiable non-monetary assets without physical form and unidentifiable future economic benefits obtained from business combinations that are not separately recognized	Number (thousand NT\$)	Output	Taiwan Economic Journal (TEJ)	Chen (2011)
	Intangible assets (IA)	Non-monetary assets that have no physical entity but have economic value and need to have three characteristics: identifiability, control and future economic benefits	Number (thousand NT\$)	Output	Taiwan Economic Journal (TEJ)	Chen (2011)
	Operating income	The balance remaining after deducting operating costs and operating expenses from operating income. Income from regular business operations, excluding sales allowances and returns	Number (thousand NT\$)	Output	Taiwan Economic Journal (TEJ)	Tone and Tsutsui (2010)
Link	Revenue (Rev)	Income from regular business operations, excluding sales allowances and returns. Identifiable non-monetary assets without physical form and unidentifiable future economic benefits obtained from business combinations that are not separately recognized	Number (thousand NT\$)		Taiwan Economic Journal (TEJ)	Chen (2011)

(Continues)

TABLE 2 | (Continued)

Stage	Factor	Definition	Numerical forms and units	Theoretical role/expected directional effect	Data source	Supporting references
Stage 2: Sustainability stage	Compensation for directors and supervisors, employee benefits (S&E benefits)	Payments to company directors and supervisors, including their regular compensation and additional benefits beyond salaries	Number (thousand NT\$)	Input	TEJ/CSR reports	Li et al. (2019); Matvos et al. (2018); Flammer and Luo (2017)
	Greenhouse gas emissions score (gas-score)	The company's score on the issue of greenhouse gas emissions in the environmental dimension (0–100) ranges from 0, being the worst, to 101, being the best	Number	Input	TEJ/CSR reports	Li et al. (2019)
	Environmental aspect score (E-score)	Assessment of environmental management practices like air and water quality, waste disposal, energy usage, and biodiversity conservation	Number	Output Link	TEJ/CSR reports	Li et al. (2019)
	Social aspect score (S-score)	Evaluation of social resource allocation, public service levels, social justice, safety, and harmony	Number	Output Link	TEJ/CSR reports	Li et al. (2019)
	Corporate governance aspect score (G-score)	Rating corporate governance structures including equity, board operations, financial reporting, risk management, CSR, and employee rights	Number	Output Link	TEJ/CSR reports	Chen and Delmas (2011)
Stage 3: Market stage	Market value	Refers to the value of an asset or a company in financial markets. It is commonly used to refer to a company's market capitalization, which is calculated by multiplying the number of shares outstanding by the current market price	Number (thousand NT\$)	Output	Taiwan Economic Journal (TEJ)	Chen and Delmas (2011)
Carry-over	Fixed assets	Physical assets used for production, rented out, or administrative purposes with a life expectancy beyond one year	Number (thousand NT\$)	Carry-over	Taiwan Economic Journal (TEJ)	Tone and Tsutsui (2010); Lu and Hung (2010)

(Continues)

TABLE 2 | (Continued)

Stage	Factor	Definition	Numerical forms and units	Theoretical role/expected directional effect	Data source	Supporting references
<i>Tobit regression variables</i>						
	Number of patents	The number of valid patents owned by a company or individual	Number	Proxy for innovation capacity and technological advancement	Taiwan patent search system	Svensson (2022)
	Enterprise scale	The size of the company in terms of various indicators, including number of employees, revenue, asset size, or market share. The enterprise scale is calculated as the logarithm of total assets	Number (thousand NT\$)	Larger scale may lead to operational and financial advantages	Taiwan Economic Journal (TEJ)	Ahmed et al. (2023)
	Age of the company	The length of time between the date of establishment of the enterprise and the current year	Number (year)	Older firms may benefit from stability, younger firms from agility	Taiwan Economic Journal (TEJ)	Clauss et al. (2019)
	Capital intensity (CIR)	An indicator that measures the degree of capital investment required by an enterprise or industry in the production process. Capital intensity is calculated as total assets divided by operating income	Number	Indicates reliance on capital investments in value creation	Taiwan Economic Journal (TEJ)	Qiao and Lu (2024); Madhani (2014)
	Debt to asset ratio	An indicator of the degree of financial leverage of a company, indicating how much of a company's assets are financed by debt. Calculated as total assets divided by total liabilities	%	Moderate leverage may enhance performance via reduced agency cost	Taiwan Economic Journal (TEJ)	Vátavu (2015)

Note: While DEA models do not assume expected directional effects, each input and output variable has been selected based on its theoretical role and empirical relevance in capturing firm performance across operational, sustainability, and market dimensions. To enhance clarity, we provide explanatory roles and literature-based justification in the revised variable table.

direction vector typically contains positive values. The scalar α indicates the magnitude of adjustment.

In the dynamic three-stage model, the entire production process is divided into three stages—S1, S2, and S3—while accounting for the characteristics of each stage in each period t ($t = 1, \dots, T$). The stages are defined as follows:

- S1 (Profit stage): Includes I inputs x_{ij}^t ($i = 1, \dots, I$) used to generate K desired outputs y_{kj}^t ($k = 1, \dots, K$). This stage is linked to Stage 2 through intermediate outputs z_{dj}^t ($d = 1, \dots, D$).
- S2 (Sustainability stage): Includes B inputs f_{bj}^t ($b = 1, \dots, B$) used to generate A desired outputs n_{aj}^t ($a = 1, \dots, A$). This stage is connected to Stage 3 through intermediate outputs h_{oj}^t ($o = 1, \dots, O$).
- S3 (Market stage): Includes G inputs E_{gj}^t ($g = 1, \dots, G$) used to generate C desired outputs q_{cj}^t ($c = 1, \dots, C$). Additionally, carry-over variables c_{hj}^t ($h = 1, \dots, H$) capture intertemporal effects.

The objective function of the DMU's efficiency is:

$$\rho_p = \max \sum_{t=1}^T \gamma_t (w_1^t \theta_1^t + w_2^t \theta_2^t + w_3^t \theta_3^t)$$

Subject to:

Stage 1: Profit stage (S1)

$$\begin{aligned} \sum_j \lambda_{j1}^t x_{ij}^t &\leq \theta_1^t x_{ip}^t \quad \forall i, \forall t \\ \sum_j \lambda_{j1}^t z_{dj(1,2)}^t &\geq \theta_1^t z_{dp(1,2)}^t \quad \forall d, \forall t \\ \sum_j \lambda_{j1}^t y_{kj}^t &\geq \theta_1^t y_{kp}^t \quad \forall k, \forall t \\ \lambda_{j1}^t &\geq 0 \quad \forall j, \forall t \end{aligned}$$

Stage 2: Sustainability stage (S2)

$$\begin{aligned} \sum_j \lambda_{j2}^t f_{bj}^t &\leq \theta_2^t f_{bp}^t \quad \forall b, \forall t \\ \sum_j \lambda_{j2}^t z_{dj(1,2)}^t &\geq \theta_2^t z_{dp(1,2)}^t \quad \forall d, \forall t \\ \sum_j \lambda_{j2}^t n_{aj}^t &\geq \theta_2^t n_{ap}^t \quad \forall a, \forall t \end{aligned}$$

$$\begin{aligned} \sum_j \lambda_{j2}^t h_{oj(2,3)}^t &\geq \theta_2^t h_{op(2,3)}^t \quad \forall o, \forall t \\ \lambda_{j2}^t &\geq 0 \quad \forall j, \forall t \end{aligned}$$

Stage 3: Market stage (S3)

$$\begin{aligned} \sum_j \lambda_{j3}^t E_{gj}^t &\leq \theta_3^t E_{gp}^t \quad \forall g, \forall t \\ \sum_j \lambda_{j3}^t h_{oj(2,3)}^t &\geq \theta_3^t h_{op(2,3)}^t \quad \forall o, \forall t \\ \sum_j \lambda_{j3}^t q_{cj}^t &\geq \theta_3^t q_{cp}^t \quad \forall c, \forall t \\ \lambda_{j3}^t &\geq 0 \quad \forall j, \forall t \end{aligned}$$

Intertemporal link:

$$\sum_{j=1}^n \lambda_j^{t-1} c_{hj}^t = \sum_{j=1}^n \lambda_j^t c_{hj}^t \quad \forall h, \forall t$$

Interstage links:

$$\begin{aligned} \sum_{j=1}^n \lambda_{j1}^t Z_{dj(1,2)}^t &= \sum_{j=1}^n \lambda_{j2}^t Z_{dj(1,2)}^t \quad \forall d, \forall t \\ \sum_{j=1}^n \lambda_{j2}^t h_{oj(2,3)}^t &= \sum_{j=1}^n \lambda_{j3}^t h_{oj(2,3)}^t \quad \forall o, \forall t \end{aligned}$$

Here, γ_t represents the weight of time t , and w_1^t , w_2^t , and w_3^t are the weights assigned to S1, S2, and S3, respectively. For all t , these weights are set to one.

Based on these equations, the following efficiency measures can be calculated using linear programming:

1. The efficiency of stage l ($l = 1, 2, 3$) for the evaluated DMU is relative to each period t ($t = 1, 2, \dots, T$).

The Stage 1 efficiency is:

$$\rho_1^t = 1 - \theta_l^{t*}; \quad l = 1, 2, 3; \quad t = 1, 2, \dots, T$$

The Stage 2 efficiency is:

$$\rho_2^t = 1 - \sum_{l=1}^T \gamma_l \theta_l^{t*}; \quad l = 1, 3$$

The Stage 3 efficiency is:

$$\rho_3^t = 1 - \sum_{l=1}^T \gamma_l \theta_l^{t*}; \quad l = 1, 2$$

2. The overall efficiency for each period t of the DMU being evaluated is expressed as follows: $\rho^t = w_1^t \theta_1^t + w_2^t \theta_2^t + w_3^t \theta_3^t$; $t = 1, 2, \dots, T$.
3. The overall efficiency across all periods is the weighted sum of periodic efficiencies, expressed as follows:

$$\rho = \sum_{t=1}^T \gamma_t \rho^t.$$

3.3 | Efficiency Determinants: Tobit Regression

The Tobit regression model is employed to examine the influence of external factors on the efficiency scores derived from the three-stage DEA model. The Tobit model is appropriate because it accounts for the censored nature of the dependent variable—efficiency scores—that are bounded between 0 and 1. This approach ensures statistical validity and provides robust estimates. The general form of the Tobit regression is:

$$E_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \epsilon_i$$

where E_i is the efficiency scores (Profit stage, Sustainability stage, or Market stage); X_{ki} is the explanatory variables (number of patents [AI index], enterprise scale, capital intensity, debt-to-asset ratio, and company age); β_k is the coefficients to be estimated; and ϵ_i is the error term, assumed to follow a normal distribution with mean zero and constant variance ($\epsilon_i \sim N(0, \sigma^2)$).

4 | Empirical Results

4.1 | Descriptive Statistics

Among the 17 variables, TSMC recorded the highest values for R&D, employees, S&E benefits, revenue, intangible assets, market value, and fixed assets across all years. Nanya led in enterprise scale, Novatek in debt-to-asset ratio, and Fintek in company age. RT had the highest Gas-score from 2020 to 2022, while TSMC dominated patent output in 2018 and Faraday from 2019 to 2022. E-scores peaked with MXIC (2018–2019) and NTC (2020–2022), whereas S-scores and G-scores were highest for TSMC (2021–2022). Minimum values included revenue and gross margin (SIS), R&D (Fintek), age (Sitronix), and CIR (SDI and Nanya). Etron recorded the lowest enterprise scale (2020–2022), and Fintek and Prolific had the lowest market values in their respective periods (Table A1).

4.2 | Efficiency Analysis

The DEA results show that EMC, TSMC, Hycon, and Chipbond achieved perfect efficiency across all 4 years, reflecting excellence in resource allocation and operational management. The average overall stage efficiency was 0.8001, the lowest among stages, indicating disparities in resource integration and performance management. Among the 33 companies, 11 fell below average, with GWC (0.1481) and Chipbond (0.0880)

ranking lowest, primarily due to poor performance in the Market stage. Leading companies consistently maintained high efficiency, demonstrating strong resource management, while low performers such as GWC and Chipbond revealed significant operational shortcomings. The overall efficiency variance (SD = 0.1903) highlights notable internal differences within the industry (Table 3).

Efficiency differences across stages are summarized in Table 4. The Market stage achieved the highest average efficiency (0.9664), followed by the Profit stage (0.9306), whereas the Sustainability stage lagged at 0.7772, with an overall efficiency of 0.8001. Thirty-one companies achieved optimal efficiency in the Market stage, reflecting robust resource allocation and reinforcing Taiwan's semiconductor industry prominence. The Profit stage also performed well, with only seven companies below average. In contrast, the Sustainability stage showed considerable room for improvement, indicating a lag in ESG integration. Annual trends reveal that the Market stage peaked in 2019, while Profit and Sustainability stage efficiencies improved until 2021. These results underscore operational strengths but highlight the need for stronger sustainability practices to support balanced growth.

In the Profit stage, 23 of the 33 DMUs achieved perfect efficiency (1.0000) across all years, reflecting effective resource allocation and management. TSMC excelled in all years except 2018 (0.8344), while VIS reached peak efficiency only in 2021. The average efficiency was 0.9306, with seven companies below this threshold, including the lowest performers: Prolific (0.7934), Etron (0.6285), Fintek (0.6062), KYEC (0.5987), and VIA (0.5965), due to uneven resource utilization. Annual efficiency improved steadily from 0.9288 in 2018 to 0.9544 in 2021 before dropping to 0.9060 in 2022, likely reflecting fluctuations in individual company performance. The efficiency gap remained relatively stable over the period (Table 5).

In the Sustainability stage, five companies (TSMC, VIA, Chipbond, Hycon, and GWC) achieved perfect efficiency (1.0000) consistently from 2018 to 2022, reflecting strong ESG practices and resource allocation. Etron and EMC excelled for 4 years, while RichWave, VIA, and Nanya demonstrated stable performance with 3 years at peak efficiency. In contrast, MTK and RT, despite strong market positions, ranked 30th and 33rd, respectively, highlighting weaknesses in ESG management. The average Sustainability stage efficiency improved to 0.7889 in 2021 but declined slightly to 0.7803 in 2022, reflecting uneven performance across companies. Overall, the average efficiency for this stage was 0.7772, suggesting considerable room for improvement. Companies such as MTK (0.5396) and RT (0.5164) need to strengthen ESG investment and management to close the gap with industry leaders (Table 6).

In the Market stage, the average efficiency was 0.9664, the highest among all stages, with 31 of 33 companies achieving perfect efficiency (1.0000) annually from 2018 to 2022. This reflects strong resource utilization and stable performance. Chipbond improved markedly after a low efficiency of 0.0096 in 2018, whereas GWC consistently underperformed, with an overall efficiency of 0.0895, indicating weak market management. The Market stage demonstrates strong integration and

TABLE 3 | Overall efficiency and period efficiency by different DMU in overall stage.

DMU	2018	2019	2020	2021	2022	2018–2023
EMC	1.0000	1.0000	1.0000	1.0000	0.9544	0.9863
TSMC	0.9448	1.0000	1.0000	1.0000	1.0000	0.9834
RichWave	1.0000	1.0000	0.9758	0.9672	0.9165	0.9578
Hycon	1.0000	1.0000	1.0000	1.0000	0.8522	0.9555
Nanya	1.0000	0.9665	1.0000	1.0000	0.8518	0.9455
SIS	0.9253	0.9329	0.9591	0.9653	0.9361	0.9156
Aspeed	0.9401	0.9392	0.9404	0.9478	0.9289	0.9089
eMamory	0.9346	0.9343	0.9264	0.9353	0.9162	0.8940
CWTC	0.8666	0.8770	0.8928	1.0000	1.0000	0.8909
Prolific	0.9487	0.9474	0.9521	0.8949	0.8617	0.8812
Weltrend	0.9155	0.9153	0.9139	0.9134	0.9234	0.8745
SDI	0.8995	0.9145	0.9350	0.9077	0.9140	0.8712
UMC	0.8410	1.0000	0.8423	0.8554	1.0000	0.8616
Generalplus	0.8997	0.9030	0.9094	0.9098	0.9039	0.8577
VIS	0.8819	0.8910	0.8604	1.0000	0.8874	0.8529
ESMT	0.8929	0.9003	0.8926	0.8506	0.8735	0.8230
WB	0.8358	0.8575	0.9108	0.9431	0.8420	0.8167
GMT	0.8901	0.8838	0.8724	0.8494	0.8784	0.8122
PXI	0.8703	0.8727	0.8704	0.8699	0.8887	0.8116
Etron	0.8727	0.8591	0.8727	0.8964	0.8710	0.8108
Fintek	0.8638	0.8604	0.8691	0.8822	0.8615	0.8008
Faraday	0.8489	0.8583	0.8524	0.9024	0.8737	0.8007
Sitronix	0.8501	0.8634	0.8462	0.8526	0.8520	0.7793
NTC	0.8424	0.8818	0.8426	0.8457	0.8406	0.7759
VIA	0.8362	0.8321	0.8457	0.8971	0.8531	0.7751
ELAN	0.8439	0.8430	0.8453	0.8439	0.8567	0.7698
MTK	0.8339	0.8529	0.8549	0.8471	0.8438	0.7698
MXIC	0.8335	0.8368	0.8437	0.8503	0.8452	0.7629
Novatek	0.8326	0.8365	0.8404	0.8475	0.8476	0.7614
RT	0.8333	0.8356	0.8400	0.8445	0.8406	0.7582
KYEC	0.6803	0.8021	0.8434	0.8565	0.8344	0.7027
GWC	0.7015	0.7059	0.7043	0.6863	0.6845	0.1481
Chipbond	0.6699	1.0000	1.0000	1.0000	1.0000	0.0880
Average	0.8736	0.8971	0.8956	0.9049	0.8859	0.8001
Min	0.6699	0.7059	0.7043	0.6863	0.6845	0.0880
Max	1.0000	1.0000	1.0000	1.0000	1.0000	0.9863
Standard	0.0813	0.0685	0.0666	0.0705	0.0629	0.1903

TABLE 4 | Overall and stages efficiency.

	Yearly efficiency					Overall efficiency
	2018	2019	2020	2021	2022	2018–2022
<i>Overall stage</i>						
Average	0.8736	0.8971	0.8956	0.9049	0.8859	0.8001
Min	0.6699	0.7059	0.7043	0.6863	0.6845	0.0880
Max	1	1	1	1	1	0.9863
SD	0.0813	0.0685	0.0666	0.0705	0.0629	0.1903
<i>Profit stage</i>						
Average	0.9288	0.9307	0.9332	0.9544	0.906	0.9306
Min	0.5086	0.4964	0.5418	0.5695	0.5565	0.5965
Max	1	1	1	1	1	1
SD	0.1528	0.1601	0.1533	0.1164	0.164	0.1367
<i>Sustainability stage</i>						
Average	0.7492	0.7872	0.7804	0.7889	0.7803	0.7772
Min	0.4977	0.5068	0.5201	0.5318	0.5217	0.5164
Max	1	1	1	1	1	1
SD	0.2078	0.1891	0.1942	0.1929	0.1909	0.1815
<i>Market stage</i>						
Average	0.9428	0.9733	0.9731	0.9715	0.9713	0.9664
Min	0.0096	0.1177	0.1129	0.059	0.0536	0.0895
Max	1	1	1	1	1	1
SD	0.2288	0.1536	0.1544	0.1638	0.1647	0.1611

stability, although underperforming firms like GWC must enhance their market strategies to align with leaders (Table 7).

4.3 | Comparison and Discussion of Efficiency Results

Based on the overall stage efficiency data, the 33 DMUs were categorized into three groups: High Efficiency (7 DMUs), Medium Efficiency (15 DMUs), and Low Efficiency (11 DMUs) (Table A2).

The High Efficiency group showed balanced resource allocation across all stages, integrating intangible assets, profitability, and ESG performance to sustain leadership. Companies such as TSMC, EMC, and Aspeed achieved complete efficiency throughout the study period (Bagna et al. 2021).

The Medium Efficiency group performed strongly in the Market stage but showed room for improvement in resource utilization and ESG performance, particularly in the Sustainability stage. Companies such as PXI and GMT maintained steady performance in the Profit and Market stages but underperformed in sustainability, underscoring the need for enhanced ESG investment.

The Low Efficiency group struggled in the Sustainability stage, with an average efficiency of 0.6844, while performing better in the Profit (0.9268) and Market stages (0.8992). A strong focus on profitability came at the expense of sustainable development, negatively affecting long-term objectives such as brand reputation and investor confidence.

A comparison in Table 8 highlights a substantial gap in intangible asset transformation between the High Efficiency group (TFE 0.8634) and the Low Efficiency group (TFE 0.6234), largely driven by weak Sustainability stage performance in the latter. High-performing companies balanced innovation, profitability, and sustainability, whereas low performers focused disproportionately on specific stages, resulting in inefficiencies. Additionally, inadequate management of intangible assets and external pressures—such as regulatory constraints and market demand fluctuations—further impeded the performance of the Low Efficiency group.

4.4 | Impact of Artificial Intelligence on Multi-Stage Efficiency

The Tobit regression results empirically validate the differentiated roles of firm-level characteristics across the three

TABLE 5 | Overall efficiency and period efficiency by different DMU in profit stage.

DMU	2018	2019	2020	2021	2022	2018–2022
UMC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
MXIC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
WB	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
SDI	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
SIS	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
RT	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Weltrend	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
MTK	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
ELAN	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
ESMT	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Novatek	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Faraday	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
PXI	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
eMemory	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
NTC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Fintek	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Aspeed	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
EMC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Chipbond	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
GWC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
CWTC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Sitronix	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
GMT	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
TSMC	0.8344	1.0000	1.0000	1.0000	1.0000	0.9669
RichWave	1.0000	1.0000	1.0000	1.0000	0.7496	0.9499
Nanya	1.0000	1.0000	1.0000	1.0000	0.7320	0.9464
Hycon	1.0000	1.0000	1.0000	1.0000	0.5565	0.9113
Prolific	0.8890	0.8856	0.8998	0.7058	0.5868	0.7934
VIA	0.6456	0.6730	0.5811	1.0000	0.6623	0.7124
Etron	0.6452	0.5772	0.6181	0.6891	0.6129	0.6285
Generalplus	0.5939	0.5824	0.6093	0.6579	0.5878	0.6062
KYEC	0.5343	0.4986	0.5418	0.5695	0.8492	0.5987
VIS	0.5086	0.4964	0.5467	0.8715	0.5593	0.5965
Average	0.9288	0.9307	0.9332	0.9544	0.9060	0.9306
Min	0.5086	0.4964	0.5418	0.5695	0.5565	0.5965
Max	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Standard	0.1528	0.1601	0.1533	0.1164	0.1640	0.1367

TABLE 6 | Overall efficiency and period efficiency by different DMUs in sustainability stage.

DMU	2018	2019	2020	2021	2022	2018–2023
TSMC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
VIS	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Chipbond	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Hycon	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
GWC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Fintek	0.9975	0.9988	0.9980	0.9888	0.9967	0.9959
Etron	0.9729	1.0000	1.0000	1.0000	1.0000	0.9946
EMC	1.0000	1.0000	1.0000	1.0000	0.8632	0.9726
Prolific	0.9571	0.9565	0.9567	0.9790	0.9983	0.9695
RichWave	1.0000	1.0000	0.9275	0.9016	1.0000	0.9658
VIA	1.0000	1.0000	0.9904	0.8199	1.0000	0.9621
Nanya	1.0000	0.8995	1.0000	1.0000	0.8234	0.9446
SIS	0.7759	0.7986	0.8772	0.8958	0.8084	0.8312
Aspeed	0.8204	0.8177	0.8211	0.8433	0.7866	0.8178
KYEC	0.5067	0.9077	0.9884	1.0000	0.6539	0.8113
eMemory	0.8037	0.8029	0.7791	0.8059	0.7485	0.7880
CWTC	0.5999	0.6309	0.6784	1.0000	1.0000	0.7818
Weltrend	0.7466	0.7458	0.7417	0.7403	0.7702	0.7489
SDI	0.6984	0.7434	0.8051	0.7232	0.7419	0.7424
UMC	0.5229	1.0000	0.5268	0.5662	1.0000	0.7232
Generalplus	0.6992	0.7091	0.7281	0.7294	0.7116	0.7155
ESMT	0.6786	0.7010	0.6779	0.5517	0.6205	0.6460
WB	0.5073	0.5726	0.7323	0.8292	0.5261	0.6335
GMT	0.6704	0.6514	0.6172	0.5481	0.6351	0.6244
PXI	0.6110	0.6182	0.6111	0.6096	0.6661	0.6232
Faraday	0.5467	0.5748	0.5571	0.7072	0.6210	0.6014
Sitronix	0.5502	0.5903	0.5386	0.5578	0.5559	0.5585
NTC	0.5273	0.6453	0.5279	0.5371	0.5218	0.5519
ELAN	0.5316	0.5289	0.5358	0.5318	0.5702	0.5397
MTK	0.5017	0.5587	0.5647	0.5414	0.5314	0.5396
MXIC	0.5004	0.5103	0.5312	0.5510	0.5357	0.5257
Novatek	0.4977	0.5095	0.5213	0.5424	0.5429	0.5228
RT	0.4999	0.5068	0.5201	0.5335	0.5217	0.5164
Average	0.7492	0.7872	0.7804	0.7889	0.7803	0.7772
Min	0.4977	0.5068	0.5201	0.5318	0.5217	0.5164
Max	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Standard	0.2078	0.1891	0.1942	0.1929	0.1909	0.1815

TABLE 7 | Overall efficiency and period efficiency by different DMU in market stage.

DMU	2018	2019	2020	2021	2022	2018–2022
UMC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
TSMC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
MXIC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
WB	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
SDI	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
SIS	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
RT	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
VIA	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Nanya	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Weltrend	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
KYEC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
MTK	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
ELAN	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
ESMT	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Novatek	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Faraday	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
PXI	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
eMemory	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
NTC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Fintek	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Generalplus	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
RichWave	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Aspeed	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
EMC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
VIS	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Etron	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Prolific	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Hycon	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
CWTC	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Sitronix	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
GMT	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Chipbond	0.0096	1.0000	1.0000	1.0000	1.0000	0.8019
GWC	0.1045	0.1177	0.1129	0.0590	0.0536	0.0895
Average	0.9428	0.9733	0.9731	0.9715	0.9713	0.9664
Min	0.0096	0.1177	0.1129	0.0590	0.0536	0.0895
Max	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Standard	0.2288	0.1536	0.1544	0.1638	0.1647	0.1611

TABLE 8 | Comparison of intangible asset TFEE between high efficiency group and low efficiency group.

		2018	2019	2020	2021	2022	Average
Group 1	TSMC	0.9300	1	1	1	1	0.9860
Group 1	EMC	1	1	1	1	1	1.0000
Group 1	Hycon	1	1	1	1	0.3869	0.8774
Group 1	RichWave	1	1	1	1	0.4081	0.8816
Group 1	Nanya	1	1	1	1	0.9974	0.9995
Group 1	SIS	0.0537	0.0828	0.1300	0.2315	1	0.2996
Group 1	Aspeed	1	1	1	1	1	1.0000
<i>Average</i>		0.8548	0.8690	0.8757	0.8902	0.8275	0.8634
Group 3	VIA	0.3858	0.2276	0.0262	0.1212	0.0611	0.1644
Group 3	Sitronix	0.3158	0.2798	0.1131	0.1241	0.1465	0.1959
Group 3	NTC	0.7102	1	1	1	1	0.9420
Group 3	ELAN	0.2964	0.4166	0.6481	1	1	0.6722
Group 3	MTK	0.2523	0.6242	0.1679	1	0.1500	0.4389
Group 3	MXIC	0.0770	0.2181	0.1106	0.2554	1	0.3322
Group 3	Novatek	1	1	1	1	0.0594	0.8119
Group 3	RT	1	1	1	1	1	1.0000
Group 3	KYEC	0.2408	0.1869	0.3592	0.5105	0.2025	0.3000
Group 3	Chipbond	1	1	1	1	1	1.0000
Group 3	GWC	1	1	1	1	1	1.0000
<i>Average</i>		0.5708	0.6321	0.5841	0.7283	0.6018	0.6234

operational efficiency stages: Profit, Sustainability, and Market. These findings reinforce the theoretical premise that heterogeneous capabilities influence firms' ability to integrate innovation and sustainability strategies effectively (Table 9).

Among the explanatory variables, the number of patents—used as a proxy for AI-driven innovations—is positively and significantly associated with efficiency in the Sustainability and Market stages and is only marginally and insignificantly associated with Overall and Profit stage efficiency. The marginal effects indicate that one additional patent increases Sustainability stage efficiency by 0.0001 and Market stage efficiency by 0.0034, while its impact on Overall and Profit stage efficiency remains economically small (0.0001) and statistically insignificant. This pattern suggests that AI-related innovation contributes more to long-term environmental and market-oriented performance than to short-term profitability. Rammer et al. (2021) highlight that AI adoption enhances innovation output, particularly in technology-intensive firms. Similarly, Svensson (2022) finds that high-value patent indicators, such as renewal duration and patent family size, strongly predict innovation success. Supporting this, Shi et al. (2023) apply a three-stage DEA model to the Chinese AI industry and confirm that AI innovation efficiency depends on both internal resources and external policy environments.

Company size (enterprise scale) is positively associated with Profit stage efficiency, suggesting that larger firms

benefit from economies of scale, better access to resources, and established operational capabilities (Ahmed et al. 2023). However, firm size is negatively and significantly associated with Sustainability and Market stage efficiency. The marginal effects further show that a one-unit increase in enterprise scale raises Profit stage efficiency by 0.0811 but reduces Sustainability and Market stage efficiency by -0.0380 and -0.5005 , respectively. This asymmetry implies that while large firms may possess the financial and technological capacity to invest in ESG initiatives and AI technologies, such advantages do not automatically translate into improved external perception or sustainable performance. The negative association may reflect challenges such as organizational complexity, slower coordination, or resistance to change, which can hinder effective ESG integration. These findings suggest that firm scale enhances financial efficiency but may constrain adaptability and responsiveness to evolving sustainability expectations.

Firm age is positively linked to Market stage efficiency, likely due to the accumulation of reputational capital, operational experience, and investor confidence (Siev and Qadan 2022). The marginal effect shows that an additional year of firm age increases Market stage efficiency by 0.0182, whereas the impacts on the Profit and Sustainability stages remain negative but statistically insignificant (-0.0019 and -0.0016 , respectively). This limited influence on other stages may indicate that legacy firms, despite their accumulated experience, face structural rigidities

TABLE 9 | Relationship between firm characteristics and efficiency at different stages.

	Overall stage			Profit stage			Sustainability stage			Market stage		
	Coefficient	Significance	Marginal effect	Coefficient	Significance	Marginal effect	Coefficient	Significance	Marginal effect	Coefficient	Significance	Marginal effect
Number of patents	0.000088	***	0.0001	0.000448		0.0001	0.000153	*	0.0001	0.010702	*	0.0034
Enterprise scale	−0.042068	***	−0.0315	0.368494	***	0.0811	−0.067913	**	−0.0380	−1.560728	***	−0.5005
Age of the company	0.001038	*	0.0008	−0.008585		−0.0019	−0.002767		−0.0016	0.056835	**	0.0182
CIR	0.000221	**	0.0002	−0.002026	**	−0.0004	0.000091		0.0001	0.578262	**	0.1854
Debt to asset ratio	0.000040		0.0000	0.127102	***	0.0280	−0.000181		−0.0001	0.001395		0.0004

Note: Asterisks indicate statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. All models control for year fixed effects.

or cultural resistance that constrain innovation adoption and ESG transformation—an interpretation consistent with organizational inertia theory.

Capital intensity exhibits divergent effects across efficiency stages. It is positively and significantly associated with Market stage efficiency, supporting Qiao and Lu (2024), who argue that infrastructure investment signals long-term commitment and enhances firm valuation. Conversely, capital intensity negatively affects Profit stage efficiency. The marginal effects confirm this asymmetry: capital intensity raises Market stage efficiency by 0.1854 but decreases Profit stage efficiency by −0.0004. Its effect on Sustainability stage efficiency is small and statistically insignificant (0.0001). This pattern suggests that while capital-intensive firms may benefit from long-term strategic positioning and stakeholder confidence, they may simultaneously suffer from rigid cost structures or suboptimal asset utilization that suppress short-term profitability.

Financial leverage, measured by the debt-to-asset ratio, is positively and statistically significantly associated with Profit stage efficiency, aligning with Park et al. (2021), who argue that moderate debt levels improve capital utilization and support productivity-enhancing investments. The marginal effect suggests that a one-unit increase in leverage raises Profit stage efficiency by 0.0280, though no significant impacts are found for Sustainability or Market stage efficiency. Its association with overall stage efficiency is negligible and statistically insignificant. This finding implies that while financial leverage may enhance short-term operational efficiency, excessive reliance on debt could elevate financial risk and reduce perceived transparency, thereby undermining stakeholder confidence in sustainability-oriented contexts.

Collectively, these findings underscore the strategic interplay between AI-driven innovation and ESG performance in resource-intensive, high-tech sectors such as semiconductors. By jointly reporting Tobit coefficients and marginal effects, the analysis highlights both the statistical significance and the substantive effect sizes of firm-level variables across different efficiency stages. Rather than serving solely as a technological tool, AI functions as a governance mechanism that facilitates sustainability transitions when embedded within coherent ESG strategies. Firms that proactively leverage AI for resource optimization, emissions monitoring, and ESG disclosure are better positioned to meet regulatory requirements and market expectations.

5 | Integrated Findings, Implications, and Future Directions

5.1 | Summary of Key Findings and Discussion

This study adopts a three-stage Data Envelopment Analysis (DEA) framework, integrated with Tobit regression, to investigate the influence of artificial intelligence (AI) and firm-specific characteristics on operational, sustainability (ESG), and market efficiency within Taiwan's semiconductor industry. The empirical results reveal notable performance asymmetries across the three stages: while efficiency in the

Profit and Market stages is generally high, the Sustainability stage consistently underperforms. This pattern aligns with prior DEA-based research in capital-intensive and high-tech sectors.

For instance, Chen et al. (2024) developed a cross-efficiency DEA model incorporating prospect theory and distance entropy to account for decision-makers' subjective preferences under risk. Their empirical analysis of high-tech industries across 29 Chinese provinces revealed substantial disparities between sustainability-related and financial efficiency metrics. Similarly, Azadi, Moghaddas, et al. (2023) and Azadi, Yousefi, et al. (2023) proposed a multiplier network analytics model with a standard set of weights to evaluate green supply chain performance. Applied to the wire and cable industry, their model demonstrated that sustainability dimensions—particularly environmental cost efficiency—often lag operational benchmarks, even under integrated network DEA frameworks. Furthermore, Badiezadeh et al. (2018) introduced a double frontier network DEA approach to assess supply chain sustainability, incorporating undesired outputs and dual-role factors. Their findings confirmed that environmental and social performance frequently trail operational efficiency, highlighting the limitations of conventional DEA models in capturing multidimensional sustainability outcomes.

These comparative benchmarks underscore the persistent structural challenges associated with ESG integration in resource-intensive contexts, even when advanced DEA methodologies are employed. The observed gap reflects the inherent difficulty of aligning digital innovation with ESG imperatives, particularly in capital- and energy-intensive industries. Contributing factors may include insufficient allocation of ESG-specific resources, regulatory ambiguity, and the complexity of embedding sustainability metrics within existing performance evaluation systems.

Patent-based AI innovation significantly contributes to sustainability and market stage efficiency, reinforcing its dual role in economic performance and environmental advancement. While patent counts are widely used as proxies for AI-driven innovation, they may not fully capture non-patented process innovations or reflect individual patents' quality and strategic relevance. To enhance methodological transparency, future studies may consider alternative AI intensity metrics—such as AI-related R&D expenditure, AI workforce share, or firm-level AI adoption surveys—which offer complementary perspectives on technological capacity and strategic orientation (Cockburn et al. 2019; Brynjolfsson and McElheran 2019).¹

These results support the work of Rammer et al. (2021) and Svensson (2022), who highlight AI's transformative potential in green manufacturing. Capelle-Blancard and Petit (2022) further argue that AI-enabled ESG analytics can enhance sustainability transparency, but only under robust institutional design. The relatively modest impact of AI on Profit stage efficiency, along with its uneven adoption across firms, suggests that technological capacity alone is insufficient. One possible explanation is that immediate financial returns in the semiconductor industry often depend on established production processes, cost control

mechanisms, and market positioning, which may not be directly influenced by AI-driven innovation in the short term. Echoing Cheng et al. (2014), effective ESG integration requires not only innovation but also strong internal governance mechanisms and strategic coherence.

Firm size and capital intensity exhibit divergent effects across efficiency dimensions. Larger firms benefit from economies of scale in profit generation; however, their operational complexity may reduce responsiveness to ESG imperatives. This finding aligns with Siev and Qadan (2022), who emphasize performance trade-offs in mature firms. Capital intensity positively influences Market stage efficiency (Qiao and Lu 2024) but may constrain short-term profitability. Financial leverage improves Profit stage efficiency (Ahmed et al. 2023), but its limited association with ESG and Market stage outcomes reflects the disconnection between economic structure and sustainability objectives.

While AI demonstrates potential in enhancing Sustainability stage outcomes, its systemic impact on ESG transformation remains constrained by organizational and institutional limitations. These constraints include fragmented ESG regulations, inconsistent data quality, and limited inter-organizational coordination, which hinder the effective deployment of AI-driven sustainability solutions. This is particularly evident among small and medium-sized enterprises (SMEs), which often face resource and capacity barriers. Moreover, as Ozkan (2024) and Svensson (2022) caution, deploying AI introduces social and ethical concerns—such as algorithmic opacity, labor displacement, and digital inequality—that may undermine broader ESG governance objectives.

The role of AI in advancing corporate governance remains a complex and underexplored dimension. As Larsson and Heintz (2020) note, algorithmic transparency and explainability are critical for integrating AI into governance structures that uphold accountability, stakeholder inclusiveness, and ethical oversight. AI has the potential to enhance ESG governance functions—such as real-time risk detection, audit automation, and compliance monitoring—but only when supported by strong institutional frameworks. Without clear safeguards, AI adoption may exacerbate information asymmetries and erode stakeholder trust.

The performance implications of AI innovation are not uniform but are conditioned by a firm's ESG readiness—that is, the maturity of its ESG frameworks and internal alignment. As Cheng et al. (2014) suggest, firms with robust ESG governance are better positioned to translate technological investments into sustainable outcomes. In this study, the limited integration of AI into ESG functions among low-efficiency firms reflects a mismatch between innovation potential and absorptive ESG capacity. These findings underscore the importance of organizational preparedness—not merely for adopting AI, but for embedding it meaningfully into sustainability-oriented strategies. Future research could explore how specific ESG governance mechanisms, such as dedicated sustainability committees or ESG-linked KPIs, mediate the relationship between AI adoption and performance.

By embedding AI and ESG within a multi-stage efficiency framework, this study contributes to the emerging literature on digital sustainability, demonstrating that AI's impact on ESG outcomes is conditional rather than automatic. Specifically, the findings advance the theoretical understanding that AI-driven innovation must be coupled with absorptive ESG capacity and governance maturity to yield meaningful sustainability benefits. This sector-specific insight offers a conceptual bridge between technological innovation and institutional readiness, highlighting the organizational mechanisms through which ESG value is realized in complex production systems.

Overall, the findings emphasize the need for targeted policy interventions, internal governance reforms, and cross-firm learning strategies to bridge the gap between technological potential and realized ESG outcomes.

5.2 | Policy and Managerial Implications

The analysis reveals that while Profit and Market stage efficiency are generally strong, Sustainability stage performance lags, particularly among SMEs. Although AI-driven innovation contributes positively to sustainability, its adoption remains uneven. Based on these findings, several targeted measures are recommended:

1. Promote scalable ESG tools for SMEs: Support the adoption of AI-based systems—such as emissions tracking and energy optimization—through cost-sharing schemes or technical assistance, enabling smaller firms to improve sustainability performance.
2. Focus R&D incentives on ESG-oriented innovation: Direct public funding toward AI applications with demonstrable environmental benefits, particularly in cleanroom efficiency and water resource management.
3. Strengthen internal ESG capabilities: Encourage firms to establish internal governance structures, data systems, and ESG execution frameworks aligned with operational objectives to close the gap between profitability and sustainability.
4. Foster cross-firm learning mechanisms: Facilitate structured knowledge exchange between high- and low-efficiency firms, particularly in ESG-AI integration.
5. Adapt ESG metrics to industry context: Develop benchmarking indicators tailored to semiconductor operations—for example, energy use per process node or AI-assisted resource recovery—to ensure relevance and comparability.

Implementing these evidence-based policy measures can help Taiwan reinforce its leadership in the semiconductor sector while advancing global sustainability objectives. Collectively, these strategies promote holistic progress across economic, social, and environmental dimensions, strengthening both industrial competitiveness and ESG leadership.

5.3 | Research Limitations and Future Directions

This study focuses on 33 publicly listed semiconductor firms in Taiwan from 2018 to 2022. While this provides insights into a critical national industry, the findings may not fully generalize to firms operating in other institutional contexts. Future research could expand the sample to include firms from other semiconductor-intensive regions, such as the United States, Japan, and South Korea, and differentiate firms by value chain positions (e.g., design, manufacturing, packaging).

The analysis relies on secondary quantitative data. Future studies may incorporate qualitative approaches—such as interviews or case studies—to capture organizational dynamics underlying ESG-AI integration more effectively. Including real-time regulatory and environmental indicators, such as carbon pricing or stakeholder ratings, could also enhance contextual relevance.

Longitudinal designs are recommended to assess the delayed impacts of innovation on ESG outcomes. Future research could examine how ESG-linked performance incentives mediate the relationship between AI adoption and Sustainability stage performance. Additionally, social externalities of AI—such as labor displacement, algorithmic fairness, and stakeholder trust—warrant further investigation, particularly within ESG governance frameworks.

5.4 | Concluding Remarks

This study proposes a three-dimensional efficiency framework grounded in the Data Envelopment Analysis (DEA) method, integrating ESG and AI indicators to evaluate multi-dimensional performance in Taiwan's semiconductor industry. Combining network-based efficiency assessment with regression analysis reveals a persistent lag in Sustainability stage performance, in contrast to consistently strong outcomes in the Profit and Market stages. This asymmetry highlights the structural and institutional challenges of embedding ESG objectives within innovation-driven, capital-intensive sectors.

The findings highlight AI's enabling role in ESG performance while emphasizing the importance of internal governance and firm-specific readiness. These insights contribute to the academic literature on sustainable innovation and inform policy efforts aimed at ESG transformation in high-tech sectors. Future research is encouraged to expand this framework to other regions and incorporate dynamic ESG mechanisms for deeper insight into the interplay between technological innovation and sustainability performance.

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Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

Endnotes

¹ For a broader discussion on AI measurement challenges and alternative metrics, see Cockburn et al. (2019) and Brynjolfsson and McElheran (2019).

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Appendix A

TABLE A1 | Descriptive statistics.

Variable	Item	2018	2019	2020	2021	2022
R&D	Ave	6,462,639	6,916,891	8,334,831	10,377,577	12,202,375
	Max	85,895,569	91,418,746	109,486,089	124,734,755	163,262,208
		TSMC	TSMC	TSMC	TSMC	TSMC
	Min	63,071	68,743.00	69,298.00	78,723.00	73,901.00
		Fintek	Fintek	Fintek	Fintek	Fintek
	SD	17,532,403	18,827,126	22,731,757	26,765,078	34,088,471
Employees	Ave	4133	4261	4666	5010	5407
	Max	48,752	51,297.00	56,831.00	65,152.00	73,090.00
		TSMC	TSMC	TSMC	TSMC	TSMC
	Min	60	60.00	60.00	76.00	78.00
		Fintek	Fintek	Fintek	Prolific	Prolific
	SD	9187	9579	10,434	11,834	13,222
S&E benefits	Ave	31,741	28,088	37,979	58,674	62,238
	Max	378,091	376,434.00	525,852.00	496,745.00	743,990.00
		TSMC	TSMC	TSMC	TSMC	TSMC
	Min	442	360.00	321.00	720.00	876.00
		VIA	Etron	Etron	Etron	RichWave
	SD	75,429	67,678	93,626	106,454	138,357
Gas-score	Ave	62.49	58.15	59.24	56.94	61.88
	Max	98.68	98.94	97.95	98.35	98.84
		Generalplus	Weltrend	RT	RT	RT
	Min	0.06	1.31	3.16	5.32	3.61
		VIS	VIS	Nanya	Nanya	VIS
	SD	33.58	33.71	31.97	34.54	32.43
Rev	Ave	58,234,767	59,461,932	73,665,520	94,835,901	117,245,787
	Max	1,031,473,557	1,069,985,448	1,339,254,811	1,587,415,037	2,263,891,292
		TSMC	TSMC	TSMC	TSMC	TSMC
	Min	223,085.00	222,952.00	160,171.00	252,153.00	182,202.00
		SIS	SIS	SIS	SIS	SIS
	SD	181,463,151	187,999,779	235,717,386	283,533,811	399,293,462
IA	Ave	610,167	790,782	1,109,975	1,078,130	1,014,010
	Max	10,862,174	14,703,688	20,165,841	21,342,644	19,888,348
		TSMC	TSMC	TSMC	TSMC	TSMC
	Min	6648.00	418.00	5764.00	4628.00	2643.00
		CWTC	Chipbond	SIS	SIS	EMC
	SD	1,929,566	2,625,649	3,642,601	3,777,464	3,529,543

(Continues)

TABLE A1 | (Continued)

Variable	Item	2018	2019	2020	2021	2022
E-score	Ave	58.53	57.55	59.71	61.08	63.83
	Max	81.55	78.66	80.90	84.74	85.53
		MXIC	MXIC	NTC	NTC	NTC
	Min	33.85	33.53	35.11	42.77	39.92
		WB	Chipbond	Chipbond	CWTC	Hycon
S-score	SD	11.19	11.72	12.20	12.47	10.70
	Ave	61.41	61.51	63.03	64.91	64.58
	Max	82.77	83.32	88.69	91.00	89.85
		UMC	UMC	Nanya	TSMC	TSMC
	Min	41.24	42.74	42.89	50.28	50.35
G-score		CWTC	CWTC	CWTC	Fintek	GMT
	SD	7.64	7.80	9.43	10.15	11.05
	Ave	56.32	56.81	58.34	59.57	60.47
	Max	75.22	75.09	77.25	76.58	75.08
		MTK	MTK	MTK	TSMC	TSMC
Gross margin	Min	37.71	37.66	43.48	38.43	39.82
		PXI	EMC	SIS	SIS	SIS
	SD	9.76	9.31	8.75	8.06	8.35
	Ave	24,397,191	23,800,999	33,092,212	45,115,616	63,026,559
	Max	497,986,041	492,698,501	711,130,120	819,537,266	1,348,354,806
Market value		TSMC	TSMC	TSMC	TSMC	TSMC
	Min	44,341.00	86,889.00	45,135.00	84,487.00	4640.00
		SIS	SIS	SIS	SIS	SIS
	SD	86,865,698	86,171,317	124,320,051	145,354,655	236,293,521
	Ave	219,383.33	326,637.48	526,574.12	645,615.24	443,934.18
Fixed-assets	Max	5,847,301	8,582,956	13,743,101	15,947,184	11,629,775
		TSMC	TSMC	TSMC	TSMC	TSMC
	Min	285.00	356.00	651.00	1922.00	1624.00
		Fintek	Fintek	Fintek	Fintek	Prolific
	SD	1,012,901	1,487,902	2,383,576	2,769,258	2,016,931
Fixed-assets	Ave	56,756,670	67,914,493	77,721,224	96,148,167	123,765,157
	Max	1,138,448,317	1,442,191,118	1,668,526,097	2,118,430,548	2,911,882,134
		TSMC	TSMC	TSMC	TSMC	TSMC
	Min	75,785	87,469	121,575	221,518	254,197
		Hycon	Hycon	Hycon	Fintek	Fintek
	SD	200,395,373	251,920,239	291,043,206	369,146,895	505,771,035

(Continues)

TABLE A1 | (Continued)

Variable	Item	2018	2019	2020	2021	2022
<i>Tobit regression</i>						
Number of patents	Ave	69.42	90.61	86.58	114.42	109.27
	Max	944.00	1333.00	1096.00	1950.00	1534.00
		TSMC	Faraday	Faraday	Faraday	Faraday
	Min	1.00	1.00	1.00	1.00	1.00
		EMC/VIS	RichWave	GWC/EMC/Etron	CWTC/KYEC	GWTC/GMT
Enterprise scale	SD	174.61	243.18	208.42	344.50	278.74
	Ave	7.05	7.09	7.16	7.28	7.29
	Max	9.32	9.36	9.44	9.53	9.65
		Nanya	Nanya	Nanya	Nanya	Nanya
	Min	5.77	5.78	5.81	5.88	5.98
Age of the company		GMT	GMT	Etron	Etron	Etron
	SD	0.83	0.84	0.83	0.84	0.86
	Ave	24.94	25.94	26.94	27.94	28.94
	Max	65.00	66.00	67.00	68.00	69.00
		Fintek	Fintek	Fintek	Fintek	Fintek
CIR	Min	7.00	8.00	9.00	10.00	11.00
		Sitronix	Sitronix	Sitronix	Sitronix	Sitronix
	SD	11.02	11.02	11.02	11.02	11.02
	Ave	23.76	24.74	27.45	23.54	21.55
	Max	423.16	428.82	497.43	417.86	338.49
Debt to asset ratio		Sitronix	Nanya	Nanya	Nanya	Nanya
	Min	0.01	0.01	0.01	0.01	0.01
		SDI	SDI	SDI	SDI	SDI
	SD	77.45	78.72	89.78	75.10	62.46
	Ave	6.03	6.72	11.73	10.72	8.66
	Max	70.97	104.98	272.33	247.40	169.00
		Novartek	Novartek	Novartek	Novartek	Novartek
	Min	1.70	1.61	1.73	1.57	1.51
		GMC	GWC	Faraday	ELAN	ELAN
	SD	11.85	17.72	46.81	42.51	28.84

TABLE A2 | Overall stage numerical range interval grouping.

Number	DMU	Sustainability stage efficiency value	Profit stage efficiency value	Market stage efficiency value	Overall stage efficiency value	Group
1	TSMC	1.0000	0.9669	1.0000	0.9834	Group 1
2	EMC	0.9726	1.0000	1.0000	0.9863	Group 1
3	Hycon	1.0000	0.9113	1.0000	0.9555	Group 1
4	RichWave	0.9658	0.9499	1.0000	0.9578	Group 1
5	Nanya	0.9446	0.9464	1.0000	0.9455	Group 1
6	SIS	0.8312	1.0000	1.0000	0.9156	Group 1
7	Aspeed	0.8178	1.0000	1.0000	0.9089	Group 1
1	eMemory	0.7880	1.0000	1.0000	0.8940	Group 2
2	CWTC	0.7818	1.0000	1.0000	0.8909	Group 2
3	Prolific	0.9695	0.7934	1.0000	0.8812	Group 2
4	Weltrend	0.7489	1.0000	1.0000	0.8745	Group 2
5	SDI	0.7424	1.0000	1.0000	0.8712	Group 2
6	UMC	0.7232	1.0000	1.0000	0.8616	Group 2
7	Generalplus	0.7155	1.0000	1.0000	0.8577	Group 2
8	VIS	1.0000	0.7124	1.0000	0.8529	Group 2
9	ESMT	0.6460	1.0000	1.0000	0.823	Group 2
10	WB	0.6335	1.0000	1.0000	0.8167	Group 2
11	GMT	0.6244	1.0000	1.0000	0.8122	Group 2
12	PXI	0.6232	1.0000	1.0000	0.8116	Group 2
13	Etron	0.9946	0.6285	1.0000	0.8108	Group 2
14	Faraday	0.6014	1.0000	1.0000	0.8007	Group 2
15	Fintek	0.9959	0.6062	1.0000	0.8008	Group 2
1	VIA	0.9621	0.5965	1.0000	0.7751	Group 3
2	Sitronix	0.5585	1.0000	1.0000	0.7793	Group 3
3	NTC	0.5519	1.0000	1.0000	0.7759	Group 3
4	ELAN	0.5397	1.0000	1.0000	0.7698	Group 3
5	MTK	0.5396	1.0000	1.0000	0.7698	Group 3
6	MXIC	0.5257	1.0000	1.0000	0.7629	Group 3
7	Novatek	0.5228	1.0000	1.0000	0.7614	Group 3
8	RT	0.5164	1.0000	1.0000	0.7582	Group 3
9	KYEC	0.8113	0.5987	1.0000	0.7027	Group 3
10	Chipbond	1.0000	1.0000	0.8019	0.0880	Group 3
11	GWC	1.0000	1.0000	0.0895	0.1481	Group 3